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Inverse compensation for hysteresis in piezoelectric actuator using an asymmetric rate-dependent model

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This paper presents a modified Bouc-Wen model for asymmetric rate-dependent hysteresis in piezoelectric actuator. On this basis, we develop a new digital inverse controller with a simple structure cascaded in the feedforward path for piezoelectric actuator. In order to eliminate modeling errors and parameter uncertainties, the developed inverse controller is combined with a feedback loop to establish a hybrid control scheme. In our experiments, the proposed model together with the developed hybrid control scheme has shown significantly reduced tracking errors caused by asymmetric rate-dependent hysteresis in piezoelectric actuator. © 2013 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4833399>]

Piezoelectric actuators are important components in scanning probe microscopes (SPMs)^{1–3} and atomic force microscopes (AFMs).^{4–6} However, because piezoelectric actuators are generally ferroelectric, they exhibit hysteretic behavior in response to an applied electric field. The non-smoothness, non-memoryless, and multi-valuedness of hysteresis pose severe limitations on the performances (tracking precision and stability, etc.) of piezoelectric actuators.^{7,8}

The linearity of piezoelectric actuator can be improved by charge control^{9,10} or model-based voltage control.^{11,12} One promising model of hysteresis is the Bouc-Wen model, originally proposed by Bouc¹³ and later generalized by Wen.¹⁴ The Bouc-Wen model features computational simplicity because it only requires one auxiliary nonlinear differential equation. Nevertheless, due to its symmetry and rate-independency, the Bouc-Wen model intrinsically contradicts hysteresis in piezoelectric actuator which is asymmetric and rate-dependent in nature.

Therefore, in order to allow the shape-control function $\psi(\dot{u}, h)$ of the Bouc-Wen model to yield independent value for each phase, we modify it as

$$\psi(\dot{u}, h) = \gamma + \phi_1 \operatorname{sgn}(\dot{u}h) + \phi_2 \operatorname{sgn}(\dot{u}) + \phi_3 \operatorname{sgn}(h), \quad (1)$$

where u is the input voltage of piezoelectric actuator; h is the hysteresis state variable; and parameters γ , ϕ_1 , ϕ_2 , and ϕ_3 are used for the adjustment of hysteresis loops. As shown in Fig. 1, $\psi(\dot{u}, h)$ has four independent values, namely $\gamma + \phi_1 + \phi_2 + \phi_3$ for $(\dot{u} \geq 0, h \geq 0)$, $\gamma - \phi_1 - \phi_2 + \phi_3$ for $(\dot{u} < 0, h \geq 0)$, $\gamma + \phi_1 - \phi_2 - \phi_3$ for $(\dot{u} < 0, h < 0)$, and $\gamma - \phi_1 + \phi_2 - \phi_3$ for $(\dot{u} \geq 0, h < 0)$. Because the value of $\psi(\dot{u}, h)$ for each phase is independent, the modified shape-control function contains four degrees of freedom (DOFs), allowing accurate modeling of asymmetric hysteresis in piezoelectric actuator.

Another disadvantage of the Bouc-Wen model is that it is essentially rate-independent. Because the hysteresis in piezoelectric actuator had been proved to be rate-dependent,^{15–18} the rate-independent property of the Bouc-Wen model does not accurately reflect the nature of piezoelectric actuator and will result in undesirable modeling errors. Therefore, in order to model the asymmetric rate-dependent hysteresis in piezoelectric actuator, we herein introduce a new parameter η to present the dynamic hysteresis property. Specifically, the asymmetric rate-dependent Bouc-Wen model for piezoelectric actuator can be constructed as

$$\dot{x} = (d_p u - h - x)/\eta, \quad (2)$$

$$\dot{h} = \dot{u}[\alpha - |h| \psi(\dot{u}, h)], \quad (3)$$

where x is the displacement of piezoelectric actuator; d_p is the piezoelectric coefficient; and α is another parameter for the adjustment of hysteresis loops. Between the extreme values of the hysteresis loops there exist certain phase lags,¹⁹ which need to be included in hysteresis modeling. Moreover, in order to avoid iteratively solving nonlinear differential equations, we transform x and h into discrete forms as

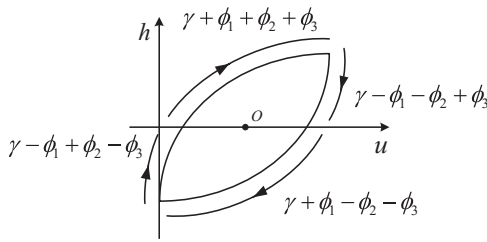
$$x_{t+1} = [(\eta - \Delta t)x_{t-\tau} + (d_p u_t - h_t)\Delta t]/\eta, \quad (4)$$

$$\begin{aligned} h_{t+1} = & h_t + \alpha(u_{t+1} - u_t) - \gamma |h_t| (u_{t+1} - u_t) \\ & - \phi_1 h_t |u_{t+1} - u_t| - \phi_2 |h_t(u_{t+1} - u_t)| \\ & - \phi_3 h_t (u_{t+1} - u_t), \end{aligned} \quad (5)$$

where the subscript t represents the parameters at time instant t ; τ is the time delay; and Δt is the sampling time.

The basic idea of inverse control strategy is to use the known response of the piezoelectric actuator to design a pre-filter for the desired displacement. Based on the inverse control scheme, hysteresis can be linearized by choosing a proper input signal. In this case, we employ a digital control system which takes a time-discrete input signal. Analytically solving u from Eqs. (1) and (3) is a difficult task. With sufficient sampling rate of time-discrete control system, we can

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FIG. 1. The asymmetric characteristic of $\psi(u, h)$.

assume $h_t = h_{t-1}$ and then h_{t-1} can be calculated according to Eq. (5). Therefore, the input signal $\tilde{u}_t$ for tracking a reference displacement $\tilde{x}_t$ can be derived as

$$\begin{aligned} \tilde{u}_t = \frac{1}{d_p} & \left[\frac{\eta(\tilde{x}_t - \tilde{x}_{t-1})}{d_p} + \tilde{x}_t + h_{t-2} + \alpha(\tilde{u}_{t-1} - \tilde{u}_{t-2}) \right. \\ & - \gamma |h_{t-2}| (\tilde{u}_{t-1} - \tilde{u}_{t-2}) - \phi_1 h_{t-2} |\tilde{u}_{t-1} - \tilde{u}_{t-2}| \\ & \left. - \phi_2 |h_{t-2}(\tilde{u}_{t-1} - \tilde{u}_{t-2})| - \phi_3 h_{t-2}(\tilde{u}_{t-1} - \tilde{u}_{t-2}) \right]. \quad (6) \end{aligned}$$

In addition, we develop a hybrid control scheme by combining the inverse controller based on asymmetric rate-dependent hysteresis model with a proportional integral differential (PID) feedback loop. The inverse controller is utilized to cancel the asymmetric rate-dependent hysteresis, whereas the PID feedback loop compensates for the modeling errors and parameter uncertainties.

Herein, an experimental platform was established for validation as illustrated in Fig. 2. A piezoceramic stack actuator (Piezomechanik GmbH) was vertically positioned such that the displacement of its top surface can be measured. A laser displacement sensor (LK-G, Keyence Corp.) with 10 nm resolution was employed to measure the output displacements of piezoelectric actuator. Signal was acquired by a dynamic signal acquisition module (PXI-4472, NI Corp.) with 24-bit resolution ADCs and 102.4 kS/s sample rate. PXI-8186 Platform (NI Corp.) equipped with LabVIEW 8.6 system was used to achieve the real-time control. The control signal was outputted by a high-speed analog output module (PXI-6733, NI Corp.) with 16-bit resolution and 10 MS/s update rate, and then amplified by a piezoelectric voltage amplifier (Rhvd-3C, RZNX Corp.) before applying to piezoelectric actuator.

TABLE I. The model parameters.

d_p (μmV^{-1})	α (μmV^{-1})	γ (V^{-1})	ϕ_1 (V^{-1})	ϕ_2 (V^{-1})	ϕ_3 (V^{-1})
0.099	0.014	-0.111	0.113	-0.112	0.097

To demonstrate the performance of the asymmetric rate-dependent Bouc-Wen model, hysteresis loops of piezoelectric actuator were measured to compare with the modeling results. In this case, the hysteresis loops were obtained by applying sinusoidal voltage with the peak-to-peak amplitude of 100 V and offset of 50 V. The model parameters are summarized in Table I. The values of η can be fitted by $\eta = 0.0074 + 0.23e^{-2.6f}$, where f is the excitation frequency. Experimental validation of the asymmetric rate-dependent Bouc-Wen model for hysteresis in piezoelectric actuator under excitations with different frequencies is shown in Fig. 3. It can be observed that the maximum hysteresis error of piezoelectric actuator increases with the frequency of applied voltage, and the corners of hysteresis loops change from sharp tips to more round tips. Moreover, the hysteresis loops turn clockwise with increasing frequency of applied voltage, resulting in a decrease of the maximum output displacement from 7.01 μm at 0.1 Hz to 6.72 μm at 10 Hz. Experimental results verify that the proposed model is competent to modeling asymmetric rate-dependent hysteresis.

Furthermore, we applied the developed control scheme to piezoelectric actuator for tracking sinusoid signals with peak-to-peak amplitude of 5 μm , offset of 2.5 μm , and frequencies of 0.1 Hz, 1 Hz, and 10 Hz, as shown in Fig. 4. Table II summarizes the tracking performance of piezoelectric actuator using the inverse controller and the hybrid controller. In addition, the compensated hysteresis loops with different frequencies are shown in Fig. 5. It can be seen that the hybrid control scheme significantly reduced the asymmetric rate-dependent hysteresis in piezoelectric actuator.

In summary, we have presented an asymmetric rate-dependent Bouc-Wen model for hysteresis in piezoelectric actuator. Experimental results demonstrate that the proposed model is effective in modeling asymmetric rate-dependent hysteresis in piezoelectric actuator. Furthermore, a new inverse controller is developed to compensate for the

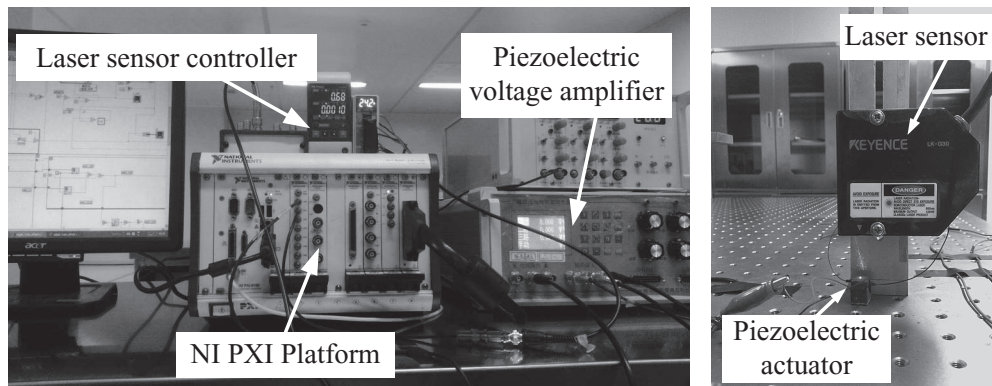


FIG. 2. The experimental platform.

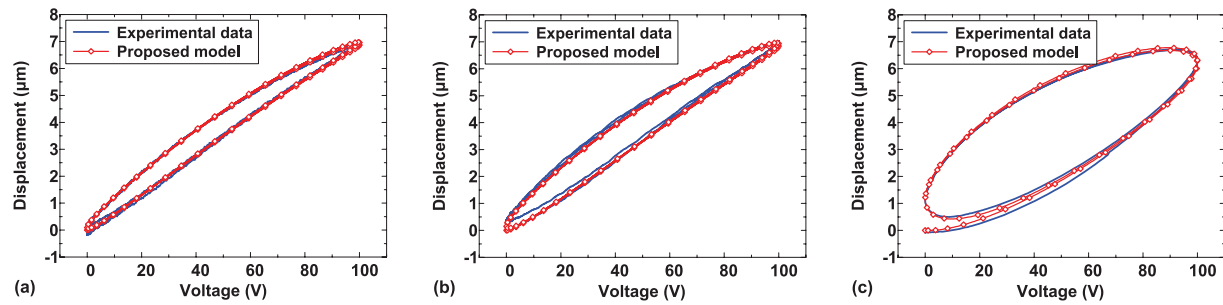


FIG. 3. Comparisons of the hysteresis loops under sinusoidal excitations with frequencies of (a) 0.1 Hz, (b) 1 Hz, and (c) 10 Hz.

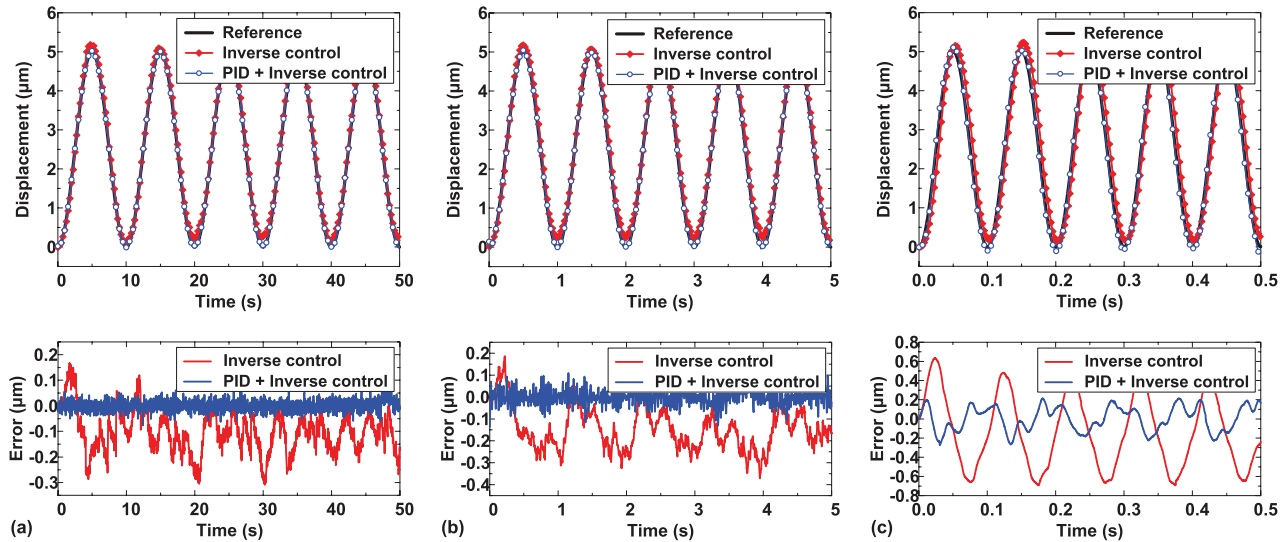


FIG. 4. Position tracking of sinusoidal waveforms with frequencies of (a) 0.1 Hz, (b) 1 Hz, and (c) 10 Hz.

TABLE II. Performance of the developed controller in tracking reference signals with different frequencies.

Frequency (Hz)	Controller	$e_{max}(\%)$	$e_{rmse}(\%)$
0.1	Inverse control	6.17	2.63
	PID + inverse control	1.71	0.42
1	Inverse control	7.42	3.51
	PID + inverse control	2.75	0.73
10	Inverse control	13.84	8.19
	PID + inverse control	5.39	2.57

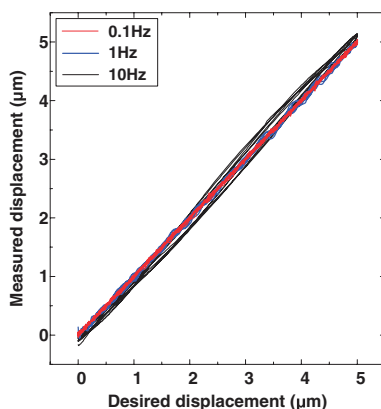


FIG. 5. Compensated hysteresis loops with different frequencies.

asymmetric rate-dependent hysteresis. In order to eliminate modeling errors and parameter uncertainties, we have combined a PID feedback loop with the inverse controller to enhance the robustness and further ameliorate the tracking performance. With the aid of this hybrid control scheme, the asymmetric rate-dependent hysteresis in piezoelectric actuator is greatly restrained.

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¹S. Moheimani and Y. K. Yong, *Rev. Sci. Instrum.* **79**, 073702 (2008).

²B. J. Kenton, A. J. Fleming, and K. K. Leang, *Rev. Sci. Instrum.* **82**, 123703 (2011).

³A. J. Fleming, S. S. Aphale, and S. Moheimani, *IEEE Trans. Nanotechnol.* **9**, 438 (2010).

⁴I. Mahmood and S. Moheimani, *Rev. Sci. Instrum.* **80**, 063705 (2009).

⁵K. K. Leang and S. Devasia, *IEEE Trans. Control Syst. Technol.* **15**, 927 (2007).

⁶D. Croft, G. Shedd, and S. Devasia, *ASME J. Dyn. Syst., Meas., Control* **123**, 35 (2001).

⁷M. Islam and R. Seethaler, *Rev. Sci. Instrum.* **83**, 116101 (2012).

- ⁸G. Y. Gu, M. J. Yang, and L. M. Zhu, *Rev. Sci. Instrum.* **83**, 065106 (2012).
- ⁹A. J. Fleming and S. R. Moheimani, *Rev. Sci. Instrum.* **76**, 073707 (2005).
- ¹⁰A. J. Fleming and K. K. Leang, *Ultramicroscopy* **108**, 1551 (2008).
- ¹¹A. A. Eielson, J. T. Gravdahl, and K. Y. Pettersen, *Rev. Sci. Instrum.* **83**, 085001 (2012).
- ¹²G. Y. Gu and L. M. Zhu, *Rev. Sci. Instrum.* **81**, 085104 (2010).
- ¹³R. Bouc, in *Proceedings of the Fourth Conference on Nonlinear Oscillation* (Academia, Prague, 1968), p. 197.
- ¹⁴Y. K. Wen, *ASCE J. Eng. Mech. Div.* **102**, 249 (1976).
- ¹⁵X. Zhang and Y. Tan, *Sens. Actuators, A* **157**, 54 (2010).
- ¹⁶P. K. Wong, Q. Xu, C. M. Vong, and H. C. Wong, *IEEE Trans. Ind. Electron.* **59**, 1988 (2012).
- ¹⁷F. Wolf, A. Sutor, S. J. Rupitsch, and R. Lerch, *Sens. Actuators, A* **172**, 245 (2011).
- ¹⁸W. Li and X. D. Chen, *Sens. Actuators, A* **199**, 89 (2013).
- ¹⁹Y. Yu, Z. Xiao, N. G. Naganathan, and R. V. Dukkipati, *Mech. Mach. Theory* **37**, 75 (2002).